

Pingchuan Ma, Ph.D.

Postdoc Research Fellow | Salk Institute for Biological Studies

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Education

Washington University in St. Louis , St. Louis, MO	Sep. 2018 – Sep. 2024
• Ph.D. in Neuroscience , McDonnell Academy Scholarship	
Harvard Medical School , Boston, MA	Aug. 2016 – Aug. 2017
• Visiting student	
Peking University , Beijing, China	Sep. 2011 -- July 2016
• Bachelor of Medicine , Merit-Based Medical Student Scholarships	

Research Experience

Postdoc Fellow , Adam Bowman Lab, Salk Institute for Biological Studies, San Diego	Jan. 2025 -- Present
• Developing new type of fluorescence lifetime imaging microscopy	
• Applying specialized optical imaging methods and molecular voltage indicators for studying neural dynamics in visual circuits	
• Combined training in optics instrumentation and neurobiology; co-advised by Adam Bowman (applied physicist) and Ed Callaway (Neurobiologist, Nation Academy of Science member)	
Graduate Student , Yao Chen Lab, Washington University in St. Louis	Sep. 2019 -- Oct. 2024
<u>Optical imaging and computational analysis for neurobiology studies</u>	
• Used <i>in vivo</i> optical imaging system and <i>ex vivo</i> two-photon microscope study neuromodulator dynamics in the brain	
• Studied brain slice physiology combining optical imaging, optogenetics, and electrophysiology	
• Utilized implants for <i>in vivo</i> bio-signal and sleep behavior assays including optical fiber and EEG/EMG implants	
• Built computational pipeline for imaging system simulation	
• Performed protein engineering of molecular biosensors	
Research Associate , X.-J. Song Lab, Southern University of Science and Technology, China	Sep. 2017 -- July 2018
<u>Chronic pain and gut-brain axis</u>	
• Studied the interactions among neurons, immune system, and gut microbiota under the context of neurological disorders	
• Utilized various disease models including nerve injury, diabetes, cancer, and chemotherapy induced chronic pain models for molecular pathogenesis of disease studies	
• Developed therapeutic targets for chronic pain including cytoskeleton and amyloid-beta in the peripheral nervous system	
Student Research Assistant , I. Chiu Lab, Harvard Medical School	Sep. 2016 -- Aug. 2017
<u>Neuroimmunology</u>	
• Studied the role of sensory neurons in host-defense of gut bacterial infections	
• Explored the impact of gut microbiota on enteric neuron functions	
Student Research Assistant , Neuroscience Research Institute, Peking University	Sep. 2014 -- July 2016
<u>Molecular biology and pain medicine</u>	
• Studied the pathogenesis of neuropathic pain in peripheral sensory neurons	

Publications

Ma, P. , Hu, L., Bowman, A. (2026). Fundamental limits on fluorescence lifetime estimation using minimally distributed temporal gates. Manuscript in preparation.	
Hu, L., Ma, P. , Menon, V., Madhavan, A. S., Asahina, K., Müller, L. M., & Bowman, A. J. (2026). Phasor EO-FLIM: Lifetime imaging with picosecond noise and 500 Hz frame rate. <i>bioRxiv</i> .	
Ma, P. , Chen, P., Sternson, S., & Chen, Y. (2025). The promise and peril of comparing fluorescence lifetime in biology revealed by simulations. <i>eLife</i> , 13, RP101559.	
Ma, P. , Chen, P., Tilden, E., Aggarwal, S., Oldenborg, A., & Chen, Y. (2024). Fast and slow: recording neuromodulator dynamics across both transient and chronic time scales. <i>Science Advances</i> , 10(8), eadi0643.	
Ma, P. , Mo, R., Liao, H., Qiu, C., Wu, G., & Song, X. J. (2022). Gut microbiota depletion by antibiotics ameliorates somatic neuropathic pain induced by nerve injury, chemotherapy, and diabetes in mice. <i>Journal of Neuroinflammation</i> , 19(1), 1-18.	
Ma, P. , Chen, P., Zhou, Z. L., Mo, R. F., Wu, M., & Song, X. J. (2020). Activation of EphB receptors contributes to primary sensory	

- neuron excitability by facilitating Ca²⁺ influx directly or through Src kinase-mediated N-methyl-D-aspartate receptor phosphorylation. *Pain*, 161(7), 1584-1596.
- Lai, N. Y., Musser, M. A., Pinho-Ribeiro, F. A., Baral, P., Jacobson, A., **Ma, P.**, ... & Chiu, I.M. Gut-Innervating Nociceptor Neurons Regulate Peyer's Patch Microfold Cells and SFB Levels to Mediate Salmonella Host Defense. *Cell*. Jan 2020.
- Chavan, S.S., **Ma, P.**, & Chiu, I.M. Neuro-immune interactions in inflammation and host defense: Implications for transplantation. *American Journal of Transplantation*. 2018, 18(3): 556-563.
- Cui, D., Li, Z. H., Li, C., Qiu, C., **Ma, P.**, Wu, M., & Song, X. J. (2022). Spinal beta-amyloid1-42 acts as an endogenous analgesic peptide in CCI-induced neuropathic pain. *European Journal of Pain*, 26(1), 133-142.
- Wu, M., Li, Z., Liang, L., **Ma, P.**, ... & Song, X. J. (2020). Wnt signaling contributes to withdrawal symptoms from opioid receptor activation induced by morphine exposure or chronic inflammation. *Pain*, 161(3), 532.
- Deng, X. T., Wu, M. Z., Xu, N., **Ma, P. C.**, & Song, X. J. (2017). Activation of ephrinB-EphB receptor signalling in rat spinal cord contributes to maintenance of diabetic neuropathic pain. *European Journal of Pain*, 21(2), 278-288; doi: 10.1002/ejp.922.
- Deng, X., **Ma, P.**, Wu, M., Liao, H., & Song, X. J. (2021). Role of Matrix Metalloproteinases in Myelin Abnormalities and Mechanical Allodynia in Rodents with Diabetic Neuropathy. *Aging and Disease*, 12(7), 1808.
- Xu, N., Wu, M.Z., Deng, X.T., **Ma, P.C.**, ..., & Song, X.J. Inhibition of YAP/TAZ activity in spinal cord suppresses neuropathic pain. *The Journal of Neuroscience*. 28 Sep 2016, 36(39): 10128-10140.
- Liu, S., Liu, Y. P., Huang, Z. J., Zhang, Y. K., Song, A. A., **Ma, P. C.**, & Song, X. J. (2015). Wnt/Ryk signaling contributes to neuropathic pain by regulating sensory neuron excitability and spinal synaptic plasticity in rats. *Pain*, 156(12), 2572-2584; doi: 10.1097/j.pain.0000000000000366.

Awards, Honors, and Memberships

- 2026, KIBM Postdoctoral Scholar Award, Kavli Institute for Brain and Mind (KIBM), UCSD
- 2025, Pioneer Fund Postdoctoral Scholar Award, Salk Institute for Biological Studies
- 2024, Excellence in Research, Division of Biology and Biomedical Sciences Washington University in St. Louis
- 2018-2023, The McDonnell Academy Scholarship, Washington University in St. Louis
- 2017-2022, Member of the Society of Neuroscience
- 2018-2020, Member of the International Association of Study of Pain
- 2017-2018, Member of the American Academy of Pain Medicine

Other Professional Development

- Leadership building

- Lo Fellow, McDonnell International Scholars Academy, Washington University in St. Louis Sep. 2018 – Sep. 2024
- Trained for leadership skills through programmed lectures, networking, and traveling
 - Initiated and led discussions of international political issues among the members of the academy
 - Invited speaker of the leadership seminar series for first-year scholars; member of the Scholar Engagement Committee

- Teaching

- Assistant Instructor, Principle of Neurobiology course, Washington University in St. Louis Sep. 2019 -- Dec. 2019

- Mentoring

- Mentored high school students, undergraduates, master students, and rotation PhD students for lab research

- Clinical training

- Intern Doctor, China-Japan Friendship Hospital, Beijing, China Feb. 2015 -- July 2015
- Clinical training and rotations in clinical departments such as surgery, orthopedics, and pain medicine

- Talks and Presentations

- 2025.11: "A theoretical framework for optical multiplexing of neuronal signals". Poster presentation, Society for Neuroscience Annual Meeting.
- 2025.10: "A theoretical framework for optical multiplexing of neuronal signals". Poster presentation, Salk Seaside Symposium.
- 2025.01: "Shining A Light on Neuroscience: New Frontiers with Fluorescent Protein-Based Sensors". Panel presentation, 2025 Winter Conference on Brain Research.
- 2024.05: "Studying the dynamics of neuromodulators in the brain by fluorescence lifetime imaging". Oral presentation, Department of Medical Neuroscience, Southern University of Science and Technology.
- 2024.04: "Beyond conventional wisdom: unveiling quantitative insights in fluorescence lifetime imaging via realistic simulation of biological systems". Poster presentation, Optics and the Brain conference (Optica).
- 2024.02: "Fast and slow: Recording neuromodulator dynamics across both transient and chronic time scales". Oral presentation, Department of Neuroscience Research Talks, Washington University in St. Louis.
- 2023.02: "Monitoring Neuromodulator Levels across Time and Animals by Fluorescence Lifetime Imaging". Oral

presentation, Biochemistry and Biophysics Community Science Friday, Washington University in St. Louis.
2022.11: "Monitoring Neuromodulator Levels across Time and Animals with Optical Sensors". Poster presentation, Society for Neuroscience Annual Meeting.
2019.11: "Gut microbiota in chronic pain caused by peripheral nerve injury, cis-platin chemotherapy, and diabetes in mice". Poster presentation, Society for Neuroscience Annual Meeting.
- **Reviewing activities:** Served as reviewer for journals *Science Advances*, *Glia*, *Communications Biology*, and *Frontiers journals*

References

- Advisor: **Adam Bowman, Ph.D.**, Salk Fellow, Salk Institute for Biological Studies, abowman@salk.edu
- Advisor: Edward Callaway, Ph.D., Professor, Salk Institute for Biological Studies, callaway@salk.edu
- Thesis advisor: **Yao Chen, Ph.D.**, Assistant professor, Department of Neuroscience, **Washington University in St. Louis**, yaochen@wustl.edu
- Thesis committee chair: **Daniel Kerschensteiner, M.D.**, Bernard Becker Professor of Ophthalmology and Visual Sciences, Vice Chair for Research in Ophthalmology and Visual Sciences, Co-Director of the Neuroscience Ph.D. Program, **Washington University in St. Louis**, kerschensteinerd@wustl.edu
- Research supervisor: **Isaac Chiu, Ph.D.**, Professor, Department of Immunology, **Harvard Medical School**, isaac_chiu@hms.harvard.edu
- Research supervisor: **Xue-Jun Song, M.D., Ph.D.**, Professor and Chair, Department of Medical Neuroscience, **Southern University of Science and Technology**, songxuejun@sustech.edu.cn